

Chapter 13 - FetchType & CascadeType

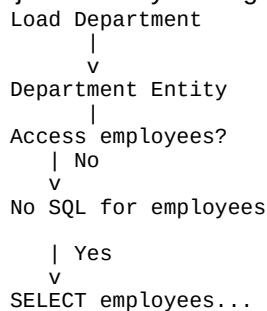
1. Introduction

FetchType controls when associated entities are loaded. CascadeType controls whether operations performed on a parent entity are propagated to related entities.

FetchType	Description
LAZY	Load associated entity only when accessed
EAGER	Load associated entity immediately with the parent

2. FetchType.LAZY

LAZY defers loading of the associated entity until it is actually accessed. Hibernate commonly uses proxy objects for lazy loading where applicable.



3. FetchType.EAGER

EAGER loads the associated entity as part of the parent entity retrieval. It may result in additional SQL or joins depending on the mapping and query.

4. Default Fetch Types

By JPA specification: • @ManyToOne -> EAGER • @OneToOne -> EAGER • @OneToMany -> LAZY • @ManyToMany -> LAZY Many projects override defaults and prefer LAZY where appropriate to improve performance.

CascadeType	Purpose
PERSIST	Propagate persist()
MERGE	Propagate merge()
REMOVE	Propagate remove()
REFRESH	Propagate refresh()
DETACH	Propagate detach()
ALL	Apply all cascade operations

5. Cascade Example

When CascadeType.PERSIST is configured on a parent-child relationship, persisting the parent automatically persists its new child entities.

```
Parent
|
persist()
|
+-----> Child 1 persisted
|
+-----> Child 2 persisted
```

6. Example Mapping

```
@OneToMany(
    mappedBy="department",
    fetch=FetchType.LAZY,
    cascade=CascadeType.ALL
)
private List<Employee> employees;
```

7. Best Practices

- Prefer LAZY loading for collections.
- Use EAGER only when the association is always required.
- Avoid CascadeType.ALL unless every cascade operation is appropriate.
- Choose cascade options based on business requirements.

8. Common Pitfalls

Accessing a LAZY association outside an active Persistence Context can result in LazyInitializationException. Design transactions and fetch strategies carefully.

9. Interview Questions

1. Difference between LAZY and EAGER? 2. What are the default fetch types? 3. Explain CascadeType.ALL. 4. When would you avoid EAGER fetching? 5. What causes LazyInitializationException?